

The d20 Annihilator: Forced Boundary Interfaces and Positive Exterior Coherence

Benjamin Huinda

Abstract

We formulate the d20 annihilator as the dual obstruction theory of a finite six-channel object. The hidden finite source is the coherent algebra $A_{985} = \text{Code}(G^{\natural})$, constructed from the Golay dodecad shell and equipped with a quotient tower $A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}$. The public d20 layer is not introduced as a numerological convention: it is forced by the six-channel support system H_6 once one requires a functorial, complement-self-dual, middle-degree boundary readout. In particular

$$d20 = \binom{H_6}{3}, \quad |d20| = \binom{6}{3} = 20.$$

The obstruction to coherent exterior passage is the positive annihilator

$$\text{Ann}_+(A_{\text{ext}}) = \{y \in \mathbb{R}_{\geq 0}^m : A_{\text{ext}}^T y = 0\},$$

where A_{ext} is the matrix of realized exterior affine circuit sides. By Gordan duality, this annihilator vanishes if and only if there is a height certificate $A_{\text{ext}} h > 0$. Thus brute-force enumeration of public d20 configurations is replaced by a finite annihilator test. The Spin_{12} and tropical layers enter through the duad chamber $1 \oplus \Lambda^2 H_6$ and the positive Pfaffian inequalities, while the d20 layer is the public middle-degree boundary $\Lambda^3 H_6$. We prove the forcing theorem, the annihilator reduction theorem, and the no-bruteforce theorem, and record a small Python certificate verifying the finite count and primitive packet layer.

1 Introduction

We begin with a finite object whose hidden multiplication is already certified: a coherent orbital algebra A_{985} with quotient tower

$$A_{985} \longrightarrow A_{236} \longrightarrow A_{42} \longrightarrow A_{12}.$$

The present paper does not re-prove the construction of this algebra. Its purpose is to isolate the boundary theorem forced by that construction. The claim is not that the number 20 is chosen because it is useful. The claim is that once the finite source has six support channels and the obstruction theory is expressed through the positive annihilator of exterior circuit systems, the minimal public self-dual boundary interface is necessarily

$$\Lambda^3 H_6, \quad \dim \Lambda^3 H_6 = 20.$$

This is the sense in which d20 is forced.

The key object is the positive annihilator

$$\text{Ann}_+(A_{\text{ext}}) = \{y \geq 0 : A_{\text{ext}}^T y = 0\}.$$

If this cone is nonzero, an exterior circuit system carries a positive dependence: a trapped transverse recirculation. If it is zero, Gordan's theorem gives a height vector h with

$$A_{\text{ext}}h > 0.$$

This converts a boundary search problem into a linear annihilator problem. In particular, one does not need to brute force the public d20 interface after the finite source and quotient tower are certified; one tests for the annihilator.

The paper is organized as follows. Section 2 records the finite source and the six-channel support. Section 3 defines exterior circuit matrices and positive annihilators. Section 4 proves the height/annihilator alternative. Section 5 proves the forcing theorem for d20. Section 6 explains the Spin_{12} and tropical chambers. Section 7 proves the no-bruteforce theorem. Section 8 connects the annihilator to regular exterior coherence. Appendix A records the accompanying proof certificate.

2 The finite source

Definition 2.1 (Finite source). *The finite source is denoted $G^{\natural}$. Its code algebra is*

$$\text{Code}(G^{\natural}) = A_{985}.$$

This algebra has a coherent orbital basis $\{R_{\alpha}\}_{\alpha=1}^{985}$ and multiplication tensor

$$R_{\alpha}R_{\beta} = \sum_{\gamma=1}^{985} p_{\alpha\beta}^{\gamma} R_{\gamma}.$$

The construction of A_{985} begins with three Hamming layers, passes through the extended binary Golay code and its dodecad shell, and uses a lifted coorient group acting on ordered pairs. The resulting quotient tower is

$$A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}.$$

The public d20 construction depends only on two consequences of this finite source:

- (i) a six-channel support system H_6 ,
- (ii) a finite quotient stack whose public layers are read through A_{42} and A_{12} .

Definition 2.2 (Six-channel support). *Let*

$$H_6 = \{B^-, B^+, V^-, V^+, S^-, S^+\}.$$

We regard H_6 as a six-element support basis. Exterior visibility data, duad coordinates, and public face packets are functorially built from this basis.

Remark 2.3. *The six labels are not used as semantic primitives in the proof. They index the six support channels inherited from the finite object. Any six-element support basis yields the same exterior algebraic statements.*

3 Exterior circuit matrices and annihilators

Let $V \subset \mathbb{R}^3$ be a finite vertex set in affine position. A five-element subset $C = \{p_0, \dots, p_4\} \subset V$ is an affine circuit when it has a unique affine dependence up to scale:

$$\sum_{i=0}^4 \alpha_i p_i = 0, \quad \sum_{i=0}^4 \alpha_i = 0,$$

with all $\alpha_i \neq 0$. The sign classes are

$$C_+ = \{i : \alpha_i > 0\}, \quad C_- = \{i : \alpha_i < 0\}.$$

The two circuit sides are

$$S_+(C) = \{C \setminus \{i\} : i \in C_+\}, \quad S_-(C) = \{C \setminus \{i\} : i \in C_-\}.$$

Definition 3.1 (Realized circuit side). *Let E be a finite exterior tetrahedron chain on the vertex set V . We say that a circuit C is realized with sign $+$ when*

$$E|_{\binom{C}{4}} = S_+(C),$$

and realized with sign $-$ when

$$E|_{\binom{C}{4}} = S_-(C).$$

If $E|_{\binom{C}{4}}$ is empty, the circuit is absent. If it is neither side, the circuit is partial and is not a row of the realized circuit matrix.

Definition 3.2 (Exterior circuit matrix). *The exterior circuit matrix A_{ext} has one row for each realized complete circuit side. If C is realized with sign $+$, the row is α_C ; if C is realized with sign $-$, the row is $-\alpha_C$.*

Definition 3.3 (Positive annihilator). *The positive annihilator of A_{ext} is*

$$\text{Ann}_+(A_{\text{ext}}) = \{y \in \mathbb{R}_{\geq 0}^m : A_{\text{ext}}^T y = 0\}.$$

A nonzero element of $\text{Ann}_+(A_{\text{ext}})$ is a positive dependence among realized exterior circuit rows.

4 Gordan duality and height certificates

Theorem 4.1 (Gordan alternative for exterior circuits). *Exactly one of the following holds:*

- (a) *there exists a height vector h such that $A_{\text{ext}} h > 0$;*
- (b) *there exists a nonzero vector $y \geq 0$ such that $A_{\text{ext}}^T y = 0$.*

Proof. This is Gordan's theorem applied to the strict linear inequality system $A_{\text{ext}} h > 0$. The alternative to strict feasibility is a nonzero nonnegative linear dependence among the rows, equivalently $A_{\text{ext}}^T y = 0$ for some $y \geq 0$, $y \neq 0$. $\square$

Corollary 4.2 (Annihilator criterion). *The exterior circuit system is height-coherent if and only if*

$$\text{Ann}_+(A_{\text{ext}}) = 0.$$

Proof. Immediate from Theorem 4.1. $\square$

The positive annihilator is therefore the precise obstruction. It is not a heuristic name. It is the dual certificate of failure for exterior coherence.

5 Why d20 is forced

The public interface must be functorially built from the six-channel support H_6 . Exterior annihilator geometry first appears in duad coordinates

$$\Lambda^2 H_6, \quad \dim \Lambda^2 H_6 = \binom{6}{2} = 15.$$

These are internal chamber coordinates, not public face states. The public interface must be complement-stable and middle-degree: if a boundary packet is present, its complement should be a packet of the same type. Complementation sends

$$\Lambda^k H_6 \longleftrightarrow \Lambda^{6-k} H_6.$$

The only nontrivial middle degree is $k = 3$.

Theorem 5.1 (d20 forcing). *Let H_6 be a six-element support basis. Assume that a public boundary interface is required to satisfy:*

- (i) *it is functorially built from H_6 ;*
- (ii) *it is complement-self-dual;*
- (iii) *it is a boundary packet layer rather than the internal duad chamber;*
- (iv) *it is minimal with these properties.*

Then the interface is

$$\Lambda^3 H_6$$

and has

$$\dim \Lambda^3 H_6 = \binom{6}{3} = 20$$

states.

Proof. The functorial exterior powers of a six-element basis are indexed by $k = 0, 1, 2, 3, 4, 5, 6$. Complementation identifies degree k with degree $6 - k$. Degrees 0 and 6 are scalar. Degrees 1 and 5 recover the channel labels and their complements. Degrees 2 and 4 are duad and co-duad layers; degree 2 is the Spin_{12} Pfaffian chamber coordinate layer. Thus the first non-scalar, non-channel, non-duad complement-self-dual degree is $k = 3$. Its dimension is $\binom{6}{3} = 20$. Therefore the minimal public boundary interface is $\Lambda^3 H_6$. $\square$

Definition 5.2 (d20 interface). *The d20 interface is*

$$d20 := \binom{H_6}{3},$$

the set of three-element subsets of H_6 . The D_6 -polarity marking supplies the legal transition structure; the state count is forced by Theorem 5.1.

Remark 5.3. *The legal transition graph is not the Johnson graph on three-subsets. It is a marked public graph obtained from the D_6 polarity/root data. The theorem above forces the public state set, while the D_6 marking supplies the legal root transitions.*

6 Spin₁₂, duads, and tropical visibility

The Spin₁₂ big-cell chart has the form

$$1 \oplus \Lambda^2 H_6.$$

The duad coordinates are

$$a_{ij}, \quad 1 \leq i < j \leq 6.$$

For each quartet $i < j < k < l$, the Pfaffian expression is

$$a_{ij}a_{kl} - a_{ik}a_{jl} + a_{il}a_{jk}.$$

The positive chamber requires these quartic Pfaffians, and the top Pfaffian, to be positive.

The tropical form uses logarithmic coordinates $L_{ij} = \log a_{ij}$. The positive tropical chamber condition is

$$L_{ik} + L_{jl} \leq \max(L_{ij} + L_{kl}, L_{il} + L_{jk}).$$

This says that the negative Pfaffian monomial is not uniquely dominant.

Proposition 6.1 (Tree visibility implies tropical positivity). *If L_{ij} is a tree metric, or more generally a valuated Pfaffian datum satisfying the tropical four-point relations, then the positive tropical Spin₁₂ inequalities hold.*

Proof. For a tree metric, the maximum of the three sums

$$L_{ij} + L_{kl}, \quad L_{ik} + L_{jl}, \quad L_{il} + L_{jk}$$

is attained at least twice. Hence the middle expression $L_{ik} + L_{jl}$ cannot be uniquely larger than both positive monomials. This is exactly the displayed inequality. The valuated Pfaffian version is the same statement written as the tropical Pfaffian relation. $\square$

7 The no-brute-force theorem

Theorem 7.1 (Annihilator reduction). *Let A_{ext} be a realized exterior circuit matrix. The obstruction to coherent exterior passage is exactly the positive annihilator*

$$\text{Ann}_+(A_{\text{ext}}) = \{y \geq 0 : A_{\text{ext}}^T y = 0\}.$$

In particular, if $\text{Ann}_+(A_{\text{ext}}) = 0$, then exterior coherence is certified by a height vector h with

$$A_{\text{ext}} h > 0.$$

Thus brute-force enumeration of public d20 configurations is replaced by the annihilator test

$$A_{\text{ext}}^T y = 0, \quad y \geq 0.$$

Proof. The first assertion is Corollary 4.2. If the annihilator is zero, Theorem 4.1 gives h with $A_{\text{ext}} h > 0$. Conversely, if such an h exists, then no nonzero $y \geq 0$ can satisfy $A_{\text{ext}}^T y = 0$, since

$$0 = h^T A_{\text{ext}}^T y = (A_{\text{ext}} h)^T y > 0$$

for nonzero $y \geq 0$, a contradiction. Hence the obstruction is exactly the positive annihilator. $\square$

8 Application to exterior coherence

Let $P \subset \mathbb{R}^3$ be weakly convex and decomposable, and let $K = \mathcal{E}(P)$ be its exterior transverse closure. The regular exterior coherence condition is

$$\exists h \quad A_{\text{ext}}(K)h > 0.$$

By Theorem 7.1, this is equivalent to

$$\text{Ann}_+(A_{\text{ext}}(K)) = 0.$$

Thus the problem of exterior coherence is exactly an annihilator problem.

In the rigidity proof, this annihilator-free condition implies positive tropical Spin_{12} visibility, hence a positive ordinary Pfaffian chamber after lifting. The local Hilbert–Einstein circuit sign then gives a positive semidefinite exterior Hessian correction. Therefore exterior coherence supplies the algebraic obstruction-theoretic content needed to replace brute-force d20 state search by a height certificate.

Theorem 8.1 (Regular exterior implication). *If a weakly convex decomposable exterior closure K satisfies*

$$\text{Ann}_+(A_{\text{ext}}(K)) = 0,$$

then the exterior data lies in the positive tropical Spin_{12} chamber and the exterior Hessian correction is positive semidefinite.

Proof. The vanishing annihilator gives a height certificate by Theorem 7.1. This gives acyclic exterior reorientation, hence tree/valuated-Pfaffian visibility. Proposition 6.1 gives the positive tropical Spin_{12} chamber. Positive leading coefficients lift to the ordinary Pfaffian chamber. The local Hilbert–Einstein circuit sign makes each positive circuit block positive semidefinite, so the exterior correction is positive semidefinite. $\square$

9 Conclusion

The d20 interface is forced because the annihilator of a six-channel finite object is read publicly in middle degree:

$$H_6 \Rightarrow \Lambda^2 H_6 \text{ annihilator chamber} \Rightarrow \Lambda^3 H_6 \text{ public boundary} \Rightarrow d20.$$

The resulting theorem is not a brute-force enumeration theorem. It is an annihilator theorem. Once the finite source and its six-channel support are certified, obstruction search is replaced by the dual test

$$A^T y = 0, \quad y \geq 0.$$

This is the finite boundary calculus underlying the d20 interface.

A Proof certificate

The accompanying proof certificate verifies the following finite identities:

$$(i) \quad \binom{6}{3} = 20, \quad \binom{6}{2} = 15, \quad \text{and} \quad 2\binom{6}{2} = 30;$$

- (ii) the complement pairing on $\binom{H_6}{3}$;
- (iii) the absence of primitive positive packet hits in the rank-two six-channel layer under one global side system;
- (iv) the dependency chain from annihilator vanishing to height certificate.

The certificate is not a replacement for the mathematical proof. It is a finite audit of the count and primitive-packet layer used by the proof.

References

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